Nptel Online Certification Course: Computer Vision
Week 11: Dimensionality Reduction and Sparse Representation
Type of questions: MCQ/MSQ
Number of questions: 10 **Total marks: 10 X 2 = 20**

QUESTION 1: **Type: MSQ**

Which of the following statements about PCA are FALSE?

a) PCA is a supervised dimensionality reduction technique.

b) The principal components are the eigenvectors of the covariance matrix.

c) PCA maximises the within-class scatter of the data.

d) The number of principal components is always equal to the number of data points.

Correct Answer: a, c, d

QUESTION 2: **Type: MSQ**

Let $X = \begin{bmatrix} 1 & 3 \\ 2 & 4 \\ 3 & 5 \\ 4 & 6 \end{bmatrix}$. Compute the unit vector along the direction of the dominant principal component of X.

a) $\begin{bmatrix} 1.0 \\ 0.0 \end{bmatrix}$

b) $\begin{bmatrix} 0.707 \\ -0.707 \end{bmatrix}$

c) $\begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$

d) $\begin{bmatrix} 0.0 \\ 1.0 \end{bmatrix}$

Correct Answer: c

Detailed Solution: Feature means: $\mu_1 = 2.5,\ \mu_2 = 4.5$.

Centred data: $X_c = \begin{bmatrix} -1.5 & -2.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$.

Covariance matrix:

$$\Sigma = \frac{1}{3} X_c^T X_c = \frac{1}{3} \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix} = \begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix}.$$

Eigenvalues: $\lambda_1 = 3.33,\ \lambda_2 = 0$.

Eigenvector for λ_1: solving $(\Sigma - \lambda_1 I)\mathbf{v} = 0$ gives $\mathbf{v} = (1,\ 1)^T$.

Normalising: $\hat{\mathbf{v}} = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$.

QUESTION 3: **Type: MSQ**

A dataset has covariance matrix eigenvalues $\lambda_1 = 50,\ \lambda_2 = 30,\ \lambda_3 = 15,\ \lambda_4 = 5$. Which of the following are true?

a) Retaining the first two principal components preserves 80% of total variance.

b) The fourth principal component contributes significantly to the total variance.

c) Retaining three components preserves exactly 95% of total variance.

d) PCA selects components to minimise reconstruction error in a least-squares sense.

Correct Answer: a, c, d

Detailed Solution: Total variance = 50 + 30 + 15 + 5 = 100.

- **(a)** First two: $\frac{50+30}{100} = 80\%$.
- **(b)** $\lambda_4 = 5$ is the smallest eigenvalue, and contributes the least to toal variance.
- **(c)** First three: $\frac{50+30+15}{100} = 95\%$.
- **(d)** PCA minimises mean squared reconstruction error; this is a fundamental property.

QUESTION 4: **Type: MCQ**

Given two class means $\mu_1 = \begin{bmatrix}1\\2\end{bmatrix}$, $\mu_2 = \begin{bmatrix}5\\4\end{bmatrix}$, and within-class scatter $S_W = \begin{bmatrix}4 & 0\\0 & 1\end{bmatrix}$, compute the optimal Fisher LDA projection direction $\mathbf{w} = S_W^{-1}(\mu_1 - \mu_2)$.

a) $\begin{bmatrix}1\\2\end{bmatrix}$

b) $\begin{bmatrix}-1\\-2\end{bmatrix}$

c) $\begin{bmatrix}-2\\-1\end{bmatrix}$

d) $\begin{bmatrix}2\\1\end{bmatrix}$

Correct Answer: b
Detailed Solution:

$$S_W^{-1} = \begin{bmatrix}\frac{1}{4} & 0\\0 & 1\end{bmatrix}, \quad \mu_1 - \mu_2 = \begin{bmatrix}1-5\\2-4\end{bmatrix} = \begin{bmatrix}-4\\-2\end{bmatrix}.$$

$$\mathbf{w} = S_W^{-1}(\mu_1 - \mu_2) = \begin{bmatrix}\frac{1}{4} & 0\\0 & 1\end{bmatrix}\begin{bmatrix}-4\\-2\end{bmatrix} = \begin{bmatrix}-1\\-2\end{bmatrix}.$$

QUESTION 5: **Type: MCQ**

Given class means $\mu_1 = \begin{bmatrix}1\\1\end{bmatrix}$, $\mu_2 = \begin{bmatrix}5\\5\end{bmatrix}$, and $S_W = \begin{bmatrix}2 & 0\\0 & 2\end{bmatrix}$, compute the Fisher LDA direction $\mathbf{w}$ and find the scalar projections of μ_1 and μ_2 onto $\mathbf{w}$.

a) $\mathbf{w} = \begin{bmatrix}-2\\-2\end{bmatrix}$; projections: -4 and -20

b) $\mathbf{w} = \begin{bmatrix}-2\\-2\end{bmatrix}$; projections: -2 and -10

c) $\mathbf{w} = \begin{bmatrix}2\\2\end{bmatrix}$; projections: 4 and 20

d) $\mathbf{w} = \begin{bmatrix}-2\\-2\end{bmatrix}$; projections: 4 and 20

Correct Answer: a
Detailed Solution:

$$S_W^{-1} = \begin{bmatrix}\frac{1}{2} & 0\\0 & \frac{1}{2}\end{bmatrix}, \quad \mu_1 - \mu_2 = \begin{bmatrix}-4\\-4\end{bmatrix}.$$

$$\mathbf{w} = S_W^{-1}(\mu_1 - \mu_2) = \begin{bmatrix}\frac{1}{2} & 0\\0 & \frac{1}{2}\end{bmatrix}\begin{bmatrix}-4\\-4\end{bmatrix} = \begin{bmatrix}-2\\-2\end{bmatrix}.$$

$$\mu_1 \cdot \mathbf{w} = (1)(-2) + (1)(-2) = -4, \quad \mu_2 \cdot \mathbf{w} = (5)(-2) + (5)(-2) = -20.$$

QUESTION 6: **Type: MCQ**

Consider a dictionary with atoms $\mathbf{d}_1 = (1,\ 1)$, $\mathbf{d}_2 = (1,\ -1)$, $\mathbf{d}_3 = (-1,\ 1)$. Using OMP with sparsity $k = 2$, find the representation of $\mathbf{y} = (4,\ 2)$.

a) $\begin{bmatrix}4\\2\end{bmatrix} = 4\begin{bmatrix}1\\1\end{bmatrix} + 2\begin{bmatrix}1\\-1\end{bmatrix}$

b) $\begin{bmatrix}4\\2\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix} + 1\begin{bmatrix}-1\\1\end{bmatrix}$

c) $\begin{bmatrix}4\\2\end{bmatrix} = 2\begin{bmatrix}1\\1\end{bmatrix} + 2\begin{bmatrix}1\\-1\end{bmatrix}$

d) $\begin{bmatrix}4\\2\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix} + 1\begin{bmatrix}1\\-1\end{bmatrix}$

Correct Answer: d

Detailed Solution: Iteration 1: Compute dot products with $\mathbf{y} = (4,\ 2)$:

$$\langle \mathbf{y}, \mathbf{d}_1 \rangle = 4 + 2 = 6$$
$$\langle \mathbf{y}, \mathbf{d}_2 \rangle = 4 - 2 = 2$$
$$\langle \mathbf{y}, \mathbf{d}_3 \rangle = -4 + 2 = -2$$

Select $\mathbf{d}_1$ (maximum absolute value = 6).
Residual: $\mathbf{r}_1 = (4, 2) - 6(1, 1) = (-2,\ -4)$.

Iteration 2: Compute dot products with $\mathbf{r}_1 = (-2,\ -4)$:

$$\langle \mathbf{r}_1, \mathbf{d}_2 \rangle = -2 + 4 = 2$$
$$\langle \mathbf{r}_1, \mathbf{d}_3 \rangle = 2 - 4 = -2$$

Either $\mathbf{d}_2$ or $\mathbf{d}_3$ (tie); select $\mathbf{d}_2$.

Least-squares solve with $A = \begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}$:

$$\begin{bmatrix}x\\y\end{bmatrix} = (A^T A)^{-1} A^T \begin{bmatrix}4\\2\end{bmatrix} = \frac{1}{4}\begin{bmatrix}2 & 2\\2 & -2\end{bmatrix}\begin{bmatrix}4\\2\end{bmatrix} \cdot \frac{1}{2} = \begin{bmatrix}3\\1\end{bmatrix}.$$

Result: $\begin{bmatrix}4\\2\end{bmatrix} = 3\begin{bmatrix}1\\1\end{bmatrix} + 1\begin{bmatrix}1\\-1\end{bmatrix}$.

QUESTION 7: **Type: MSQ**

Which of the following are true about sparse representation and dictionaries?

a) A dictionary can be over-complete, meaning it has more atoms than the signal dimension.

b) OMP is a greedy algorithm that selects one atom per iteration.

c) Sparse representation requires all atoms of the dictionary to be used.

d) Basis Pursuit replaces the ℓ_0 minimisation with an ℓ_1 minimisation for tractability.

Correct Answer: a, b, d

Detailed Solution: Over-complete dictionaries have more columns than rows, providing richer representational capacity. OMP greedily selects the atom most correlated with the current residual at each iteration. Sparse representation by definition uses only a *small* subset of the dictionary atoms; the representation coefficient vector has few non-zero entries. The ℓ_0 norm (count of non-zeros) minimisation is NP-hard; Basis Pursuit relaxes it to ℓ_1 minimisation, which is convex.

QUESTION 8: **Type: MCQ**

In the K-SVD algorithm, during the **dictionary update** step for atom $\mathbf{d}_j$, which of the following correctly describes the procedure?

a) Replace $\mathbf{d}_j$ with the mean of all data points assigned to it.

b) Compute the residual error matrix $E_j = X - \sum_{i \neq j} \mathbf{d}_i \mathbf{y}_i^T$, then apply rank-1 SVD to update $\mathbf{d}_j$ and the corresponding coefficients.

c) Re-run OMP on the entire dataset and replace $\mathbf{d}_j$ with the dominant eigenvector.

d) Set $\mathbf{d}_j$ to the normalised sum of all training signals.

Correct Answer: b

Detailed Solution: In K-SVD, the dictionary atom $\mathbf{d}_j$ is updated one at a time.
Let ω_j be the set of indices of training signals that use atom j.
The residual error matrix for atom j is:

$$E_j = X - \sum_{i \neq j} \mathbf{d}_i \mathbf{y}_i^T,$$

where $\mathbf{y}_i^T$ is the i-th row of the sparse coding matrix restricted to ω_j.
The rank-1 SVD of E_j gives $E_j \approx \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T$.
The updated atom is $\mathbf{d}_j \leftarrow \mathbf{u}_1$ and the updated sparse coefficients are $\sigma_1 \mathbf{v}_1$.
This ensures $\|X - DY\|_F^2$ is non-increasing after each atom update.

QUESTION 9: **Type: MCQ**

Which of the following correctly identifies how K-SVD generalises K-means clustering?

a) K-SVD uses ℓ_2 norm for the reconstruction error, while K-means uses ℓ_1 norm.

b) In K-means each data point is represented by exactly one cluster centre; in K-SVD each signal can be represented by a sparse combination of multiple dictionary atoms.

c) K-SVD initialises its dictionary randomly, whereas K-means always uses a fixed initialisation.

d) K-means allows each data point to be represented by multiple cluster centres; K-SVD restricts representation to one atom.

Correct Answer: b

Detailed Solution: K-means assigns each point to a *single* centroid (sparsity $k = 1$, binary coefficients). K-SVD generalises this by allowing each signal to be reconstructed from a sparse linear combination of $k > 1$ dictionary atoms with real-valued coefficients.

QUESTION 10: **Type: MCQ**

Consider two classes: $X_1 = \{(2,\ 4),\ (4,\ 2),\ (3,\ 3)\}$ and $X_2 = \{(8,\ 6),\ (6,\ 8),\ (7,\ 7)\}$. Compute the within-class scatter matrix S_W.

a) $S_W = \begin{bmatrix} 2.0 & -2.0 \\ -2.0 & 2.0 \end{bmatrix}$

b) $S_W = \begin{bmatrix} 1.33 & -1.33 \\ -1.33 & 1.33 \end{bmatrix}$

c) $S_W = \begin{bmatrix} 4.0 & 0.0 \\ 0.0 & 4.0 \end{bmatrix}$

d) None of the above

Correct Answer: b

Detailed Solution: Class 1 mean: $\mu_1 = (3,\ 3)^T$.
Class 1 scatter:

$$S_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}\begin{bmatrix} -1 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}.$$

Class 2 mean: $\mu_2 = (7,\ 7)^T$.
Class 2 scatter:

$$S_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}\begin{bmatrix} 1 & -1 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix}\begin{bmatrix} -1 & 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix}\begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}.$$

Within-class scatter:

$$S_W = \frac{S_1 + S_2}{3} = \frac{1}{3}\begin{bmatrix} 4 & -4 \\ -4 & 4 \end{bmatrix} = \begin{bmatrix} 1.33 & -1.33 \\ -1.33 & 1.33 \end{bmatrix}.$$
